

Carrier and Timing Recovery for QPSK Communication

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1. Problem Statement

In a practical QPSK communication system, the receiver does not have perfect knowledge of the carrier phase, carrier frequency, or exact symbol timing. Channel noise, oscillator mismatch, frequency offset, phase offset, and sampling-clock differences can cause constellation rotation and incorrect symbol decisions. These synchronization errors increase the Bit Error Rate (BER) and reduce communication reliability. Therefore, an efficient carrier and timing synchronization system is required before demodulation.

2. System to be Built

The proposed system is a software-based QPSK communication receiver with carrier and timing synchronization. Random binary data will be QPSK modulated, pulse-shaped, and transmitted through a simulated channel containing noise, carrier frequency/phase offset, and timing error. At the receiver, matched filtering, timing recovery, and carrier recovery will synchronize the signal before QPSK demodulation. Constellation diagrams and BER measurements will be used for evaluation.

System Flow:

Random Bits → QPSK Modulator → RRC Pulse Shaping → Noisy Channel + Synchronization Errors → Matched Filter → Timing Recovery → Carrier Recovery → QPSK Demodulator → Recovered Bits → BER & Constellation Analysis

3. Expected Results

The system is expected to recover QPSK data reliably in the presence of noise and synchronization errors. Expected outcomes include successful symbol-timing recovery, correction of carrier frequency and phase errors, improved constellation alignment, reduced bit errors, lower BER, BER-versus-SNR analysis, and a clear comparison of system performance before and after synchronization.

4. Tools and Software

Software: MATLAB, Communications Toolbox, DSP System Toolbox, and Signal Processing Toolbox.

Optional: Python with NumPy, SciPy and Matplotlib

Future Hardware: USRP or RTL-SDR may be used to extend the simulation to real-time communication.

Conclusion

The proposed project demonstrates how carrier and timing synchronization improve the reliability of a QPSK receiver. By combining synchronization algorithms with BER and constellation analysis, the project provides practical insight into digital communication receiver design and DSP-based synchronization.